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13 AUGUST 2026

DEMONSTRATE AN UNDERSTANDING OF TESTING IT SYSTEMS AGAINST GIVEN SPECIFICATIONS

US 14908

SECTION B: SUMMATIVE ASSESSMENT

TASK 1

1a. Testing Specifications

Specifications received from Supervisor:

SYSTEM: Student Registration Software v1.2

DATE: 13 August 2026

SPECIFICATIONS TO TEST:

1. Users must be able to login with username and password. Login time < 3 seconds.

2. System must not allow duplicate Student ID numbers to be captured.

3. All student personal information must be stored in line with POPIA.

4. System must generate student registration reports in PDF format.

5. System must support minimum 50 concurrent users without crashing.

Approved by: _____ Date: _____

Supervisor

Fikile Mkhize

1b. IT Systems Testing Plan

IT Systems Testing Plan

- Project Name: Student Registration Software v1.2
- Tester: Fikile Mkhize
- Date of Test: 13 August 2026

- Purpose: To verify that the software meets all functional, security and performance specifications.
- Scope: Login Module, Student Registration Module, Search Module, Report Generation Module.
- Test Approach: Functional Testing, Security Testing, Performance Testing
- Test Environment: Dell PC, Windows 10, 8GB RAM, Test Database Copy
- Resources Required: Test data sheet, Test cases, Stopwatch, Screen capture tool
- Roles & Responsibilities: Tester = , Supervisor = Witness and Sign-off
- Schedule: 09:00 – 12:00
- Risks: Risk of using live data. Mitigation: Only use test data.
- Compliance: POPIA, Copyright Act, OHSA, Organisational IT Policy
- Pass/Fail Criteria: Minimum 90% of test cases must pass. No critical defects.

1c. Test Scenarios and Test Data

Test ID	Scenario	Test Data
TS01	Valid user login	Username: admin, Password: Admin123
TS02	Invalid password	Username: admin, Password: 12345
TS03	Add new student record	ID: 2026001, Name: Thabo Mokoena, DOB: 12/03/2005
TS04	Capture duplicate Student ID	ID: 2026001, Name: Lindiwe Dlamini
TS05	Search for existing student	Search: 2026001
TS06	Generate registration report	Date range: 01/08/2026 to 16/08/2026

1d. Expected Outcomes

Test ID	Expected Outcome
TS01	Login successful. Dashboard loads in less than 3 seconds.
TS02	Error message: “Invalid username or password” is displayed.
TS03	Student record saved. Confirmation message: “Student added successfully”

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TS04 Error message: “Student ID already exists” is displayed. Record not saved.

TS05 Correct student details for ID 2026001 are displayed.

TS06 PDF report is generated and downloads with correct data.

Task 2

Report: Selection of Test Procedure

Purpose of testing: To confirm that the Student Registration Software functions according to specifications and complies with POPIA and organisational standards.

Factors considered:

1. Time: Testing must be completed in 3 hours.
2. Budget: No additional software to be purchased. Use existing PC.
3. Type of system: Desktop application handling personal data.
4. Skills: Tester has training in functional testing.

Resources required:

Test PC, Test user accounts, Test database, Supervisor for observation.

Compliance requirements:

1. POPIA: Ensure personal data is not exposed during testing.
2. Copyright Act: Ensure software license is valid
3. OHSA: Ensure workstation setup is ergonomic and safe.
4. Organisational policy: Follow change management and backup procedures.

TASK 3

Supervisor Evaluation Checklist:

- [ ] Test plan was followed
- [ ] Hardware and software tested correctly
- [ ] Performance parameters verified
- [ ] Health and safety procedures followed
- [ ] POPIA and licensing requirements met
- [ ] Problems identified and action taken

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Supervisor comments:

All procedures were met and everything was done accordingly.

Signature: _____

Date: 13/08/2026

Task 4

4a. TEST REPORT

1. Introduction

Testing was conducted on 16 August 2026 on the Student Registration Software v1.2 to verify compliance with specifications.

2. Method

Functional Testing and Security Testing were applied using the test scenarios in Task 1.

3. Results

Test ID	Expected Outcome	Actual Outcome	Pass/Fail
TS01	Login <3s	Login 2.1s	Pass
TS02	Error message	Error message. displayed.	Pass
TS03	Student saved	Student saved	Pass
TS04	Duplicate error	Duplicate error displayed	Pass
TS05	Correct details	Correct details shown	Pass
TS06	PDF generated	PDF took 9 seconds	Fail

4. Defects Identified

DEF01: Report generation takes 9 seconds. Specification requires less than 5 seconds. Severity: Medium.

5. Conclusion and Recommendations

5 out of 6 tests passed. The software is fit for use with the condition that DEF01 is logged for developer to optimize report generation. No data security issues found. System complies with POPIA.

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6. Sign-off

Tester Name: Fikile Mkhize Signature_____

Date: 13/08/2026

Supervisor Name: _____

Signature: _____

Date: 13/08/2026